

Question:

Which of the following factors is associated with a decreased risk of Alzheimer's?

Answer Choices:

- A. Being African or Hispanic American
- B. A lower level of education
- C. Living in a city
- D. Being married
- E. Having a family history of Alzheimer's
- F. High intake of saturated fats
- G. Older age
- H. Eating fish
- I. Being a smoker
- J. Drinking alcohol regularly

Reasoning + Answer:

< reasoning process here >

To determine which factor is associated with a decreased risk of Alzheimer's, we need to evaluate each option based on current medical and epidemiological research.

A. Being African or Hispanic American → Studies have shown that African American and Hispanic individuals may have higher risks due to socioeconomic, health access, and genetic factors, so this is not protective.

B. A lower level of education → Lower educational attainment is associated with a higher risk of Alzheimer's, likely due to reduced cognitive reserve. So this is not protective.

C. Living in a city → Urban living has mixed associations, but some studies suggest urban environments may be associated with higher stress or pollution, not lower risk.

D. Being married → Research has consistently shown that marital status can be protective; married individuals tend to have lower rates of dementia, possibly due to social support, reduced loneliness, and behavioral factors. This is a well-documented protective factor.

E. Having a family history of Alzheimer's → Family history is a known risk factor, so it is not protective.

F. High intake of saturated fats → High intake of saturated fats is linked to increased risk of cardiovascular disease and neurodegeneration, so it is not protective.

G. Older age → Age is a major risk factor for Alzheimer's; older age directly increases risk. So not protective.

H. Eating fish → Fish, especially fatty fish like salmon, are rich in omega-3 fatty acids and are associated with a reduced risk of dementia. This is supported by multiple studies.

I. Being a smoker → Smoking is a known risk factor for Alzheimer's, so not protective.

J. Drinking alcohol regularly → Excessive drinking is harmful; however, moderate alcohol intake (especially in certain contexts) has been linked to a lower risk, particularly red wine and its antioxidants. But "regularly" is ambiguous → high or low frequency matters. Still, moderate intake is linked to benefits, but regular drinking (without moderation) may not be protective.

Now, among the options, both D (being married) and H (eating fish) are associated with reduced Alzheimer's risk.

However, the question asks: "Which of the following factors is associated with a decreased risk?"

We must pick the one most clearly and consistently supported by evidence among the options.

Among all, the evidence for fish consumption (especially omega-3s) and marital status is strong and consistent in neuroprotection. However, eating fish is a dietary factor, and the evidence is strong and direct.

But let's reassess: being married is a well-documented social protective factor (based on longitudinal studies like the one from the Journal of the American Medical Association). Eating fish is also protective. Both could be correct, but the question only allows one answer.

Looking at the options again, the most robust and widely cited protective factor from social/environmental factors is marriage.

However, omega-3 from fish is also well-documented in reducing neurodegeneration.

But in medical literature, factors such as social engagement, physical activity, and diet (like fish) are protective. Among the options, eating fish is a positive dietary habit.

But consider: "Having a family history" is known risk, not protective. Similarly, "older age", "smoking", and "low education" increase risk.

Between D and H, which is better supported?

Multiple large studies (e.g., the Nun Study, Framingham) show that married people live longer and have lower rates of dementia. Marital status is strongly linked with reduced risk.

Fish consumption has also been associated with reduced risk of Alzheimer's, especially when rich in omega-3s.

But the most strongly supported and consistent protective factor not just in diet, but in social structure, is being married.

Moreover, the options include "being married", which is a social protective factor and that is one of the most well-established risk-reducing factors.

So, between D and H, which is best?

The answer "H. Eating fish" is correct → many studies show that diets rich in omega-3

